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Exploring the factors influencing undergraduate students' decision to pursue physics at the University of Education, Winneba

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ABSTRACT

Background: Over the years, the number of prospective students intending to study physics at the University of Education, Winneba (UEW) has gradually declined. This has drastically affected the number of trained physics teachers in the country. As a matter of urgency, there is therefore the need to look at the factors that influence the choice of physics at the undergraduate level in UEW in order to encourage more students to enrol in the course.

Purpose: This study sought to find out the factors motivating the students at UEW, to major in physics, despite the general belief that the course is very difficult.

Sample: The participants of this study were physics students at the Department of Physics Education, UEW, Ghana. Purposive sampling was used to select 101 respondents.

Design and Methods: Questionnaire was used in gathering data. Interview and focused group discussion were used to validate the responses of the participants to the questionnaire. The questionnaire was analyzed using R version 4.22. Additionally, Relative Importance Index (RII) was used to rank the Likert form items.

Results: The findings indicated that students ranked interest (RII = 0.92) as their major motivating factor to read physics in the Department. Students also claimed that their parents had little influence on their decision to study physics. Whiles professional teachers used it as a conduit to secure study leave with pay from GES. Students who had difficulty in the subject also attributed it to their long delay at home after completing senior high school before gaining admission to the University.

Conclusions: The study recommends that there should be a concerted effort to arouse the interest of physics learners at the lower levels of education in Ghana by the Department of Physics Education, Winneba and other stakeholders in order to increase the enrolment of students in physics.

KEYWORDS

Interest; choice of physics; study leave; professional teachers; job prospects

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Introduction and background

The study of physics is important to humans in order to understand our world both around us, inside us and the world beyond us (Hapgood et al. 2021). Physics is also the most basic and fundamental natural science involving universal laws and the study of the behaviour and relationships among a wide range of important physical phenomena (Yahaya, Gimba, and Gana 2020). Moreover, it includes the investigation of the entire cosmos, ranging from the biggest galaxies to the tiniest subatomic particles. Physics contributes to the technological infrastructure and provides trained personnel needed to take advantage of scientific advances and discoveries. It also improves our quality of life by providing the basic understanding necessary for developing new instrumentation and techniques for medical applications, such as computer tomography, magnetic resonance imaging, positron emission tomography and ultrasonic imaging among others. It therefore serves as the foundation for other disciplines such as biology, chemistry and engineering.

Despite the important role that physics plays in shaping the world, it remains the least favoured science course among students globally (Erinosho 2013). Comparatively with other science courses, only few students usually choose to read physics at higher levels. It is worth noting that the desire to pursue physics at the undergraduate level is largely influenced by the success rate and the foundation a student receives in physics at senior high school (Opare et al. 2018). Unfortunately, most senior high-school physics teachers teach physics in an abstract manner, hence, making it difficult for students to grasp basic concepts (Coffie, Frempong, and Appiah 2020). Some students are compelled to adapt to rote learning and memorization in order to pass their examinations. Those who are unable to adapt to rote learning may develop, apathy, fear and hatred for the subject. The inability of students to conceptually understand basic concepts in physics has resulted in the poor performance of students in physics over the years (WAEC 2022). Such a predicament is deemed undesirable for the developmental trajectory of a country, particularly one in the process of growth, such as Ghana, and must be addressed (Antwi 2013).

Physicists and science educators have conducted various researches to explore potential solutions to this issue. For instance, Buabeng and Ntow (2010) revealed a wide range of reasons, which accounted for students' negative response to physics in Ghana. Prominent among these reasons were; teacher factor, poor performance, perceived difficult nature of physics and unknown career opportunities in the subject. Additionally, some factors influencing poor performance in physics are; inadequate number of teachers, lack of teaching and learning materials, poor teaching methods and students' attitudes towards science subjects (Majo 2018). Consequently, this has resulted in declining trend in the enrolment of students into physics programmes in the various institutions in Ghana.

In the case of University of Education, Winneba, the enrolment of students into physics programme has consistently been declining over the years. Records indicate that, out of 149 students admitted into various programmes in the Faculty of Science Education as far as in the 2003/2004 academic year, only 11 (7.3%) opted to read physics (Taale 2011). Currently, out of the 1,729 students admitted into the Faculty of Science Education in the 2019/2020 academic year, only 89 (5.1%) opted for physics. This continues to decline in the subsequent years with 70 (4.1%) in 2020/2021 academic year, 51 (3.1%) in 2021/2022 academic year and 35 (2.1%) in 2022/2023 academic year. Whereas, the total population

for the university increased from 38,827 in 2020/2021 academic year to 39,196 in the 2022/2023 academic year (Table 1).

This trend is not only peculiar to this University, but also prevalent in other universities in Ghana. For instance, in the 2018/2019 academic year, only 18 of the 680 students enrolled in Physical Sciences at the University of Cape Coast chose to study physics. This figure dropped to 16 out of 746 students admitted to study Physical Science in the 2019/2020 academic year (Vice Chancellor’s Report 2020).

Meanwhile these two universities; University of Education, Winneba and the University of Cape Coast, are mandated to train teachers to teach physics in senior high schools in Ghana. The dearth of physics majors at the undergraduate level would undoubtedly have a debilitating effect on the supply of in-field teachers for the nation’s high schools and other physics related jobs, as witnessed in the UK (Holgate 2023).

On the other hand, there has been a significant increase in the number of students enrolled in Ghana’s senior high schools since the Free Senior High School Policy was implemented in 2017. The population of enrolled students rose from 260,210, in 2017, to 424,092 in 2018; about 72 % increment (MoE 2018). However, this rapid expansion has exacerbated pre-existing challenges, particularly the critical shortage of qualified science teachers, with physics instructors being disproportionately scarce (Adu-Gyamfi et al. 2020).

To mitigate this gap, many schools have resorted to assigning out-of-field teachers to physics classrooms (Buabeng, Ossei-Anto, and Ampiah 2014) a practice that has compromised educational quality. Consequently, while more students now pursue science education, few achieve passing grades in physics, and even successful candidates often lack foundational conceptual understanding (Antwi et al. 2021). This deficit has contributed to a growing reluctance among students to pursue physics at the tertiary level, undermining the policy’s broader goals.

Addressing this decline in university-level physics enrollment requires a nuanced understanding of the factors shaping students’ academic choices. Research highlights multifaceted influences, including societal perceptions of science, parental guidance, and personal motivations. For instance, Lipps (1999) emphasizes the role of societal recognition of science’s practical value in shaping student aspirations, while Venville et al. (2013) identify parental influence as a key driver. Similarly, Mujtaba and Reiss (2014) outline a spectrum of motivators, such as extrinsic material rewards, intrinsic interest in physics, self-concept in the subject, and classroom experiences. Collectively, these academic, social, and personal dimensions underscore the complexity of students’ decision-making processes, offering critical insights for policymakers seeking to revitalize physics education and align it with Ghana’s educational ambitions.

Table 1. Enrolment statistics of physics students in UEW.

Academic Year	Number of students opting for Physics	Total number of students admitted into Faculty of Science Education	Full-Time Undergraduate Students’ Enrolment of UEW
2019/2020	89 (5.1%)	1729	49,468
2020/2021	70 (4.1%)	1692	38,827
2021/2022	51 (3.1%)	1667	38,931
2022/2023	35 (2.1%)	1635	39,196

(Source: UEW 28th Congregation Basic Statistics, 2024)

Some students chose to study physics because of the ease in obtaining study leave with pay (Osei-Gyasi 2010). While others chose to study physics because it has better future aspirations for them (Fullarton and Ainley 2000; Osborne and Collins 2001). Moreover, as noted by Taale (2010), students at the University of Education, Winneba, often chose to pursue physical science because it fostered a strong sense of self-concept and self-efficacy in mathematics. In addition, some students reported that their confidence to undertake the course stemmed from encouragement received through informal interactions with physics lecturers (Bolshakova, Johnson, and Czerniak 2011; Garza and Landeck 2004; Mujtaba and Reiss 2014). However, it is also important to acknowledge the existence of a group of students who decides to study physics even though it was not their initial choice of discipline (Holmegaard, Madsen, and Ulriksen 2014; Meli et al. 2020).

Contrastingly, the narrative of low enrolment in physics observed in some contexts does not hold in the case of Greece, where a relatively high number of students opt for physics (Meli et al. 2020; Tzanakou 2017). According to the literature, factors contributing to this trend include favourable job prospects and a genuine interest in the subject (Jones and Trippenbach 2016; Regan and Dewitt 2014). Furthermore, Meli et al. (2020), drawing on the work of Hazari et al. (2010), suggest that career-related motivations and the intrinsic desire to engage with physics for intellectual fulfilment may also account for the higher enrolment figures. These tendencies are further shaped by Greece's unique socioeconomic context. Physics programs in Greece are widely regarded as being designed for high-achieving students, due to their rigorous academic requirements and esteemed institutional reputations. This observation aligns with a broader, cross-cultural narrative that positions science as an elite domain accessible primarily to those deemed intellectually exceptional, an idea supported by findings from Australia and New Zealand (Venville et al. 2013) and echoed by students in the United States (Masnick et al. 2010).

Notwithstanding the above factors on both low and high enrolment of students in these countries, Ndalichako (2014) is still of the opinion that most science students in the secondary schools made decisions regarding the choice of science programmes with inadequate information. Most often, their minds are occupied with the perception that the subject is difficult and abstract (Checkley 2010). Ornek, Robinson, and Haugan (2008) identified three categories of factors that contribute to make students see physics as difficult. These include factors which students were capable of controlling, those related to the course and factors inherent in the nature of physics. One other study identified emotional issues as the largest contributing factor to students' perception of physics (Bray and Williams 2020). The study also revealed that there was a high correlation between study skills, and students' understanding and problem-solving skills.

Many of these studies were carried out in other countries. It is therefore important to investigate why physics students in Ghana particularly University of Education, Winneba chose to do physics. The motivations behind these students' decision to major in physics might aid in dispelling the perceptions surrounding the discipline and encourage more students to enrol in physics courses.

For this study, Holland's theory (Holland 1959), rational choice theory (Hastie and Dawes 2009), social cognitive career theory (Lent, Brown, and Hackett 1994) and Krumboltz's Planned Happenstance Theory (Krumboltz 2009) have been adopted to serve as conceptual frameworks for examining individuals' career decisions from multiple perspectives.

Holland's theory is widely utilized in school career counselling (Ramadhani, Jannah, and Putri 2020). It categorizes individuals into five personality types: realistic, investigative, artistic, social, enterprising, and conventional (RIASEC). By identifying their personality type, individuals can select careers that align with their interests. Holland emphasized that greater congruence between a person's interests and their academic or career choices leads to higher satisfaction and success (Rocconi, Liu, and Pike 2020).

The strength of the RIASEC model lies in its structured categorization of interests and personality traits into six distinct types, accommodating students from diverse backgrounds. It has been empirically validated in numerous studies (Armstrong et al. 2008; Hogan 1982) and is effective in predicting outcomes in educational and vocational contexts (De Cassia Nakano and Primi 2024).

While RIASEC remains a straightforward tool for understanding student choices in educational settings, some literature points out its limitations (Zainudin et al. 2024) such as its oversimplification of complex human behavior by fitting diverse interests into a fixed six-category framework (Panakaje et al. 2024). Additionally, while the model establishes general patterns of person-environment fit, it may overlook non-rational influences such as emotions and social factors (Arnold 2004), as well as the impact of incomplete information on decision-making, particularly for senior high-school graduates in Ghana. Therefore, careful integration with other models is necessary to address the complexities of human decision-making processes.

Rational choice theory of education on the other hand, posits that students make choices by weighing the costs and benefits of each option to maximize personal utility (Gabay-Egozi, Shavit, and Yaish 2010). In the context of student choices at the University of Education, Winneba (UEW), students may therefore choose to study physics when they are convinced that the benefits such as strong career opportunities, study leave with pay, scholarships, and promising job placements outweigh perceived costs, including the subject's difficulty and potential expenses. By assuming that students, like all decision-makers, aim to maximize their benefits, rational choice theory facilitates comparisons among academic options.

This theory has widespread use across various social sciences, including decision-making in psychology and economics (Mellers, Schwartz, and Cooke 1998) and subject selection among disciplines such as physics, chemistry, and social sciences (Gabay-Egozi, Shavit, and Yaish 2010). However, the theory has faced criticism for its unrealistic assumptions, lack of empirical testing, and limited validity (Green and Shapiro 1994; Ogu 2013). It assumes that students have complete and accurate information when making decisions, which is often not the case, especially at the secondary school level (Ndalichako 2014). Furthermore, while rational choice theory emphasizes economic factors, many students are motivated by non-economic considerations such as personal fulfilment and social impact – that the theory may not fully capture.

Complementing the limitations of Holland's RIASEC and rational choice theories are Social Cognitive Career Theory and Krumboltz's Planned Happenstance Theory (Krumboltz 2009). Both theories, derived from Bandura's work (Bandura 1986), emphasize the role of self-efficacy and outcome expectations in shaping career decisions (Lent and Brown 2019).

According to Social Cognitive Career Theory, a student's decision to pursue physics is influenced not only by rational cost-benefit analysis but also by their belief in their ability

to succeed and the anticipated outcomes of their actions. Encouragement from family, teachers, or role models can also sway student preferences, highlighting the impact of social influences alongside rationality.

Additionally, some students may select physics by chance, as it was the program offered to them by the university, even if it was not their first choice. Krumboltz's Planned Happenstance Theory suggests that individuals may encounter unplanned events that influence their career paths (Chen and Lappano 2023). Experiences gained through such circumstances may spark interest and ultimately lead to a career in physics, even if it was not the student's initial intention.

Together, these theories acknowledge the importance of individual preferences in career decision-making, illustrating that students weigh factors such as personal interests, career prospects, self-efficacy, and perceived benefits when making educational choices.

The study seeks to answer two critical research questions:

- (1) What factors significantly influence undergraduate students' decisions to pursue a degree in physics at UEW?
- (2) How do demographic variables moderate the impact of these factors on students' choice to study physics?

Methodology

To answer the research questions, this study employs a mixed-methods approach, combining surveys and interviews to gather data from undergraduate students enrolled in the physics program. The research questions guided the design of the survey and interview protocols, ensuring that data collection focused on key themes related to student demographic variables, intrinsic and extrinsic factors, institutional factors, and challenges faced in physics education. By structuring the methodology around these questions, the study ensures a comprehensive analysis of the factors influencing students' choices. The mixed-method design used in this study was to ensure rapid turnaround of the data collection, reduce cost and increase the anonymity of participants. It also allows for the collection and analysis of both qualitative and quantitative data concurrently, thereby enabling a more comprehensive understanding of a research problem (Creswell and Clark 2017)

The population was physics students at UEW who had studied physics for at least one (1) academic year at the time of the study. This was done in order to explore the personal experiences of students in their study of physics. The population of the students was 210. Of these, 51 were in level 200, 70 in level 300 and 89 in level 400. Students in the first year were not included because they had done only one semester at the time.

Based on accessibility and voluntary participation, a total of 101 undergraduate physics students responded to a questionnaire via google form. The selection of this sampling method was driven by its practicality, as it facilitated straightforward access to the students with the exception of the level 400 students. The level 400 students, at the time of the study, were undergoing their internship programme in senior high schools all over the country. Consequently, a substantial proportion of these students could not participate in the study. The non-participation may be attributed to various factors, including limited access to the internet, among other considerations. Simple random

method was used to select 10 students for a personal interview and 5 students each from levels, 200, 300 and 400 for a focus group discussion.

The questionnaire was designed from reviewed literature and our interactions with students in the Department. The items on the questionnaire contained questions on demographics, year of completion of senior high school, grade obtained in Elective Mathematics (E-Maths), whether participants were professional teachers or not, and whether participants regretted their choice of physics. Thirty-five percent of the questions were structured in the Likert form based on a scale of 'strongly disagree to strongly agree' (Link to questionnaire: <https://forms.gle/3MsVb32Bkpqqrn8n6>).

To ascertain the content validity of the questionnaire, a comprehensive literature review was conducted to identify all pertinent factors influencing the decision-making process of prospective students contemplating a pursuit of a physics degree. Prominent among them were the ASPIREs projects (Ker et al. 2013), Hazari et al. (2010), Mujtaba and Reiss (2013), Taale (2010), Mahmud and Adnan (2023), Erinosho (2013), etc. Additionally, a set of open-ended questions on reasons why students choose to do physics was administered to some selected students and responses collated, analysed and grouped into themes such as interest, difficulty of course, role of role models and family members, available option offered by the university, job prospect, among others. These themes were in line with the factors reviewed in the literature. Subsequently, the questionnaire underwent rigorous scrutiny by experts in the field of physics education, and their feedback was duly integrated into its refinement. Reliability of the questionnaire items was assessed utilizing the Cronbach's alpha test, yielding a coefficient of 0.846, indicative of satisfactory internal consistency.

Prior to its implementation, the questionnaire underwent a pilot phase within a comparable academic institution, where its suitability and relevance for the present study were affirmed.

Some items from the questionnaire were selected for incorporation into both interview and focus group discussion protocols. The veracity of the interview data was ensured through meticulous transcription and subsequent member checking procedures. Interviews were conducted by a pair of researchers duly trained for this purpose.

Each participant was approached for consent, wherein the objectives and significance of the study were comprehensively elucidated. Additionally, participants were explicitly informed of their autonomy to engage or withdraw from the study at their discretion.

The focus group discourse was administered by a triad of students who underwent requisite training for the facilitation of this activity. This procedural step was undertaken to foster an environment conducive to uninhibited participant expression, thereby mitigating concerns regarding potential reprisal or victimization. The principal student facilitators were furnished with a comprehensive questionnaire delineating the inquiries to be posed to their peers. Furthermore, participants engaged in the focus group dialogue were thoroughly briefed on the overarching objectives of the study and affirmed their entitlement to voluntary withdrawal from participation at any juncture. Prior to the commencement of audio recording, participants provided explicit consent for the documentation of proceedings.

The statistical software R version 4.22 was utilized to compute percentages and means, which were subsequently employed for result interpretation. The transcripts of interviews and focused group discussions were transcribed and juxtaposed with responses obtained

from the questionnaire. Additionally, the Relative Importance Index (RII) methodology was employed to rank the factors influencing the decision to pursue studies in physics according to their significance.

The RII formula, as adapted from Khatib, Poh, and El-Shafie (2020), is represented as follows:

$$RII = \frac{\sum w}{AN} = \frac{1n_1 + 2n_2 + 3n_3 + 4n_4 + 5n_5}{5N}; (0 \leq RII \leq 1) \quad (1)$$

where n_1, n_2, n_3, n_4, n_5 denote the number of respondents indicating varying degrees of agreement, ranging from 'strongly disagree' to 'strongly agree', respectively. The coefficients (1, 2, 3, 4, 5) correspondingly denote the scales for each level of agreement. Furthermore, N represents the total number of respondents involved in the study.

Demographics of students

There were 101 students that took part in the study, and most of them were males (Table 2). Participants' ages ranged from 20 to 39 years old, with a mean age of 28 years. The majority of the students (52.5%) belonged to the 20–29 age range. The minimum and maximum ages of the males and females were 20 and 21 years old, respectively, and 39 and 36 years old, for the males and females, respectively. When the ages of the participants were compared to the ages of some students at the University of Cape Coast (Edjah, Domey, and Ankomah 2019), it was observed that participants' ages did not differ from those at the University of Cape Coast. Thus, this age range of students, may not be unique to the University of Education, Winneba.

Some of the participant had spent a maximum of about 20 years before entering the university. This could be due to reasons such as financial constraints, health issues, change of career path, inability to secure study leave from work place, among others.

The average years students spent after SHS before gaining admission into the university did not vary much for both males and females (7.13 and 7.15 years respectively).

Additionally, 57% of the participants were professional teachers; 50 males and 8 females. In accordance with the prevailing standards in Ghana, a professional teacher is defined as an individual who has undergone formal training in pedagogy from a duly

Table 2. Demographic data of students.

No.	Respondent Background	Category	Count	Percentage (%)
1	Gender	Female	13	12.9
		Male	88	87.1
2	Age	20–24	30	29.7
		25–29	23	22.8
		30–34	35	34.7
		35 – above	13	12.9
3	Professional Status	Professional Teachers	58	57.4
		Non Teachers	43	42.6
4	Years after completing SHS	1–3	39	38.6
		4–6	2	2.0
		7–9	21	20.8
		10–13	34	33.7
		14 – above	5	5.0
	Total		101	100

accredited institution within the country. Notably, teacher-training programmes in Ghana are predominantly offered either through colleges of education or universities.

In the context of senior high-school instruction, prospective educators are mandated to possess a university-level degree in education, specializing in their chosen academic discipline, such as a Bachelor of Education (B.Ed) in Physics, Home Economics, Biology, among others. Colleges of education, on the other hand, cater to the preparation of educators primarily for the foundational levels of education encompassing Kindergarten, primary, and junior high schools. Graduates from these institutions are thus recognized as professional teachers within the Ghanaian educational framework (Buabeng, Ntow, and Otami 2020).

Furthermore, incumbent upon basic school educators is the prerequisite to fulfil a minimum tenure of 3 years of teaching service subsequent to graduation, prior to becoming eligible for study leave with pay. This provision allows them the opportunity to pursue further academic endeavours, typically at the university level.

Results and discussion

This section presents the results and discussion in relation to the research questions, drawing on both quantitative and qualitative data. The findings are derived from students’ responses to specific questionnaire items such as ‘I chose to study physics because I wanted a challenging course’ and ‘My parents influenced my decision to study physics’, among others as shown in Table 3, and are further supported by interview data. The discussion examines the diverse factors influencing students’ academic choices, emphasizing the complex interplay between intrinsic and extrinsic motivations, as well as socio-demographic characteristics.

I chose to read physics because I needed a difficult course to read

The percentage of students who strongly disagreed with the statement was 23.8%, 31.7% of them agreed with the statement and 17.8% of the students neither agreed nor disagreed with the statement. The number of students who at least disagreed with the

Table 3. Factors influencing students’ choice of physics at UEW.

Factors influencing the choice of Physics at University of Education, Winneba	Likert Scale					RII	Rank
	Strongly disagree	Disagree	Not sure	Agree	Strongly agree		
I chose to read physics because I needed a difficult course	24 (23.8%)	16 (15.8%)	18 (17.8%)	32 (31.7%)	11 (10.9%)	0.58	4th
My parents/benefactor asked me to study physics	32 (31.7%)	20 (19.8%)	13 (12.9%)	30 (29.7%)	6 (5.9%)	0.52	5th
My role model encouraged me to read physics	19 (18.8%)	16 (15.8%)	8 (7.9%)	29 (28.7%)	29 (28.7%)	0.67	3rd
I chose to read physics because I had interest in the course	2 (5.0%)	0 (0.0%)	6 (5.9%)	22 (21.8%)	71 (70.3%)	0.92	1st
I chose to read physics because it was the only option available for me to choose	35 (34.7%)	19 (18.8%)	14 (13.9%)	28 (27.7%)	5 (5.0%)	0.50	6th
I chose to read physics because it has better job prospect	6 (5.9%)	11 (10.9%)	7 (6.9%)	37 (36.6%)	40 (39.6%)	0.79	2nd
I did not choose physics but the University offered me admission to read physics	48 (47.5%)	8 (7.9%)	13 (12.9%)	30 (29.7%)	2 (2.0%)	0.46	7th

statement was 40 (39.6%) and those who at least agreed with the statement was 43 (42.6%). This implies that many of the physics students were aware of the inherent difficulty in studying the course but were prepared to challenge themselves in overcoming those challenges. It is important to note that majority of the females (53%) agreed to this statement as compared to the males (38.5%).

A cross tabulation of age group with the statement (Table 4) revealed that within the age group, 33.3% of the youngest age group (20–24 years) disagreed with the statement, 47.8% (25–29 years), 42.9% (30–35 years), and 30.8% (those above 34 years) disagreed with the statement. Within the age groups, the percentages that agreed with the statement were : 46.7% (20–24 years), 43.5% (25–29 years), 40.0% (30–34 years) and 30.8% (35 years and above). When compared with the other factors using the relative importance index (RII), this factor ranked 4th among the 8 factors (Table 4). It is quite interesting to note that the desire to study physics because of its inherent difficulty waned with increase in the ages of the participants. The younger generation appeared to have chosen to read physics because of its perceived difficult nature than the older generation.

Interestingly, this suggests that the challenging nature of the subject serves as a motivating factor for certain individuals rather than dissuading them. This phenomenon aligns with the concept of Desirable Difficulty (DD) effect posited by Bjork and Bjork (2011), which suggests that engaging with more challenging material necessitates greater cognitive exertion from learners, thereby fostering enhanced learning outcomes. For these students, their primary objective is to delve deeper into the intricacies of the subject's complexity, aiming to demonstrate the feasibility of mastering physics to sceptics.

Remarkably, a notable proportion of female participants in the study cited the perceived difficulty of physics as a driving force behind their decision to pursue the discipline. Moreover, societal perceptions often portray science as a domain reserved exclusively for individuals of exceptional intelligence, further amplifying the significance of accomplished physicists as role models for aspiring young students within the community (Lipps 1999). Comparable findings from studies conducted in Australia and the United States (Masnick et al. 2010; Venville et al. 2013) corroborate this notion, indicating a global resonance of the phenomenon. For these students, the pursuit of physics not only contributes to their academic credentials but also enhances their social standing and prestige among peers, particularly among younger cohorts who perceive physics as a vehicle for elevating their status within social circles.

Table 4. Comparison of age and the choice of a difficult course.

		Age group				Total
		20–24	25–29	30–34	35 – above	
I chose to read physics because I needed a difficult course	Strongly disagree	6	3	11	4	24
	Disagree	4	8	4	0	16
	Not sure	6	2	6	4	18
	Agree	10	6	11	5	32
	Strongly agree	4	4	3	0	11
Total		30	23	35	13	101

My parents influenced me to study physics

For this statement, 31.7% of the students strongly disagreed, 19.8% disagreed, 29.7% agreed and 5.6% strongly agreed. Therefore, the number of students who claimed that their decision to study physics was independent of their parents' influence was 52 (51.5 %). The decision of majority of the students to read physics seems to have had insignificant parental influence.

When parental influence was cross tabulated with age group (Table 5), it was observed that 50% of students aged 20–24 years, 52.2% (25–29 years), 54.2% (30–34 years) and 46.2% (more than 34 years) disagreed with the statement. From this result, it appears parental influence on the students' choice of physics decreased with increasing age. Thus, the older the students grew, the less the influence of their parents on their choice of subjects. This is not surprising, since majority of the respondents for this study were mature enough (above 25 years) and had the ability to make right choices for themselves.

The revelation regarding minimal parental influence on students' selection of physics resonates with the findings outlined by Olaosebikan and Olusakin (2014), who investigated the impact of parental influence on the career choices of adolescents in the Badagry Local Government Area of Lagos State, Nigeria. Their study revealed that approximately 48% of participants acknowledged the influence of parents on their career decisions, with only 22% expressing willingness to pursue careers aligned with their family's profession. Conversely, a significant majority, comprising 79% of respondents, expressed a lack of inclination toward continuing family traditions in their chosen professions. Consequently, the researchers concluded that the career decisions of young adults within the studied schools appear to be largely independent of parental influence.

Similarly, Dewitt et al. (2013) and Taale (2011) corroborate the findings of this study, suggesting that there is minimal parental influence on students' choice of physics.

However, it must be acknowledged that the influence of parents on their children's career choices can manifest both covertly and overtly (Parola and Marcionetti 2022; Venville et al. 2013). Kumar (2016), Vautero, Silva, and Do Céu Taveira (2021), Mahmud and Adnan (2023), as well as Siddiky and Akter (2021), have all identified a significant parental influence on their children's career pathways. Such disparities in findings may be attributable to variances in socio-cultural contexts and other contextual factors (Taale 2011). Moreover, the majority of students in this study were professional teachers and significantly older in age. As adult learners, their academic decisions were likely influenced by a different set of factors, with parental influence playing a minimal role.

Table 5. The influence of parents/role models on students age and gender to choose physics.

	My parents asked me to study physics					My role model encouraged me to read physics		
	Age group					Gender		
	20–24	25–29	30–34	35 – above	Total	Male	Female	Total
Strongly disagree	10	4	13	5	32	16	3	19
Disagree	5	8	6	1	20	12	4	16
Not sure	4	1	6	2	13	8	0	8
Agree	9	9	9	3	30	25	4	29
Strongly agree	2	1	1	2	6	27	2	29
Total	30	23	35	13	101	88	13	101

My role model encouraged me to read physics

Majority of the students (57.4 %) were influenced by their role models to study physics (Table 3). This means that, role models especially, physics teachers, have a greater role to play in reversing the decline in the number of students opting to read physics at the university. Physics teachers are therefore encouraged to make their physics lessons interesting and practical.

Also, 31.8% of males and 53.8% of females claimed they were not influenced by their role models (Table 5). Similarly, 59% of males and 46.2% of females were encouraged by their role models to read physics. More males than females were inspired by their role models to read physics. As articulated by Mujtaba and Reiss (2013), girls were found to be statistically significantly less probable to receive encouragement from their teachers compared to their male counterparts. This observation is unsurprising, given the disproportionate representation of male physics instructors relative to their female counterparts in Ghanaian senior high schools (Opare et al. 2018). Additionally, the age group that was most influenced by role models was 25–29 years and followed by those above 34 years. In all, there was over 50% influence of role models on the students to study physics.

The significance of mentorship and educator motivation warrants considerable attention in the context of academic decision-making processes. Hence, the impact of role models emerged as a significant factor influencing the subject selection of students. Notably, non-professional teachers, particularly those of younger age cohorts in comparison to their professional teacher counterparts, exhibited a heightened susceptibility to the influence of role models in their decision to enrol in the physics programme.

As posited by Garza and Landeck (2004), the interpersonal milieu of these students, encompassing their interactions with teachers, lecturers, and esteemed scientists whom they regard as influential figures, assumed a pivotal role in fostering academic motivation and perseverance among students.

I chose to read physics because I had interest in the course

Interestingly, about 92% of the students were reading physics because of their interest in the course (Table 3). Only 2% did not like the course. The high interest of the students in physics would be enough motivation for them to overcome any challenge that may hinder their study of physics. Interest therefore seems to be the motivational factor for students opting to do physics. It ranked first with relative importance index of 0.92 among the other factors. Over 95% of males and about 85% of females' students said that their choice of physics emanated from their interest in the course (Table 6).

The high interest of students in physics cuts across all the groupings such as: age group (Table 7), professionalism and number of years at home. Interest therefore

Table 6. The interest to read physics and students' gender.

Gender	I chose to read physics because I had interest in the course					Total
	Strongly disagree	Disagree	Not sure	Agree	Strongly agree	
Male	2	0	4	19	63	88
Female	0	0	2	3	8	13
Total	2	0	6	22	71	101

Table 7. The interest to read physics and their age.

		Age group				Total
		20–24	25–29	30–34	35 – above	
I chose to read physics because I had interest in the course	Strongly disagree	1	0	1	0	2
	Not sure	4	0	2	0	6
	Agree	10	3	7	2	22
	Strongly agree	15	20	25	11	71
Total		30	23	35	13	101

emerged as a pivotal determinant influencing students' choice to pursue physics studies at the University of Education, Winneba (UEW), according to the findings of this investigation. This discovery aligns with the conclusions drawn by Jones and Trippenbach (2016) in their examination encompassing students from 18 European nations, wherein personal interest was identified as a predominant motivating factor in selecting physics as a field of study. Also, research suggests that physics interest is a significant predictor of academic achievement and persistence in science-related fields (Regan and Dewitt 2014). It plays an important role in encouraging students to enrol and remain in any science programme.

Notably, a gender disparity was observed, with male students exhibiting a greater inclination towards physics compared to their female counterparts as observed by Mujtaba and Reiss (2013). This observation also corroborates existing literature indicating a prevailing gender imbalance within the field of physics, with females typically displaying lower levels of interest relative to males (Morgan and Aboagye 2022). Moreover, it is imperative to advocate for the inclusion and encouragement of female students in physics education, recognizing the potential for such initiatives to enhance their interest in the subject matter. As highlighted by Hazari et al. (2010), in an era where no academic degree can guarantee unequivocal success, the pursuit of one's passion holds paramount significance. Despite prevalent misconceptions regarding the perceived difficulty of physics, among other deterrents, a considerable number of students still exhibit a genuine interest in the discipline. When students harbour a profound interest in a subject, they are intrinsically motivated to engage ardently in their academic endeavours, thereby fostering a conducive environment for academic achievement.

Given that the University undertakes the responsibility of preparing these students for roles as educators in secondary schools, their intrinsic interest in the subject assumes heightened importance. Consequently, educators should endeavour to cultivate students' innate curiosity and enthusiasm for physics through the provision of stimulating learning experiences, including practical demonstrations, collaborative projects, and innovative pedagogical approaches. Recognizing the pivotal role played by the high school level in shaping students' aspirations and foundations for tertiary education, secondary school physics instructors bear a significant responsibility in making the subject matter engaging and accessible to students (Opare et al. 2018). If we intend to increase the number of physicists in the Country, then, we need to work hard at increasing the interest of basic and second cycle students in physics.

I chose to read physics because it was the only option available to me

The response of the students shows that 53.5% of them disagreed with this statement, 32.7% agreed and 13.9% were indecisive (Table 3). This means that majority of the students had other options but opted for physics. Also, out of the 88 males, 53.4% disagreed with the statement and 34.1% agreed with the statement. The number of females who disagreed with the statement was 7 (53.8%) and those that agreed was 4 (30.8%). Almost the same percentage of males (53.4%) and females (53.8%) were of the opinion that they chose physics not because it was the only option available to them but due to other factors (Table 8). This seems to be the least agreed choice for pursuing physics in the University of Education, Winneba, except for the professional teachers, where 19 out of the 58 said they opted for physics because that was the only course that could grant them study leave. When those 19 students were asked whether they had any regrets studying physics, only six of them responded in the affirmative. Three of the 6 students who regretted studying physics gave the following reasons;

Student A: I stayed in the house for long so I forgot most of the concepts.

Student B: I don't get the marks that I want no matter how hard I try.

Student C: The course wasn't the way I perceived it to be, especially at the University level.

In examining the duration preceding admission to the university for these 6 students, it was found that 3 individuals spent 10 years, 1 individual spent 11 years, and the remaining 2 individuals spent 12 years following their completion of senior high school. This analysis sheds light on the varied pathways undertaken by students prior to their enrolment, illustrating the diverse trajectories leading to university admission.

Concerning the propensity of students to select alternative courses in lieu of physics if presented with the opportunity, a prevailing sentiment of reluctance was expressed among the majority of respondents. This sentiment persisted even among respondents occupying professional teaching roles, with only a minority expressing a theoretical preference for alternative courses, subject to the condition of receiving study leave with pay from the Ghana Education Service (GES). Despite potential influences such as coercion or external pressures impacting their decision to pursue the physics programme, respondents generally conveyed contentment with their academic pursuits within the Department. This assertion underscores the satisfaction derived from their chosen field of study despite potential considerations of alternative paths.

Surprisingly, the prevalence of regret among students who had initially chosen physics was minimal. Among those expressing regret, the primary attributions centred on the

Table 8. Gender of students and physics being the only option for them to choose.

Gender	I chose to read physics because it was the only option available for me to choose					Total
	Strongly disagree	Disagree	Not sure	Agree	Strongly agree	
Male	30	17	12	24	5	88
Female	5	2	2	4	0	13
Total	35	19	14	28	5	101

prolonged period of inactivity prior to matriculation at the University of Education, Winneba (UEW), with an average duration of 11 years. This phenomenon aligns with the principles espoused by Thorndike's Law of Disuse, positing that learned behaviours may diminish or fade from memory in the absence of sustained practice or engagement (Thorndike 1913).

Consequently, this underscores the imperative for lecturers within the Department of Physics Education at UEW to proactively investigate and address the unique challenges faced by their students. By garnering a comprehensive understanding of these challenges, educators can tailor their instructional methodologies to foster a conducive learning environment, thereby promoting effective teaching and learning outcomes.

I chose to read physics because it has better job prospects

On job prospects, only 5.9% of the students strongly disagreed with the statement, 10.9% disagreed, 36.6% agreed and 39.6% strongly agreed with the statement (Table 3). This means that 76.2% of the students are studying physics because of the better job prospects they are likely to benefit from, while only 16.8% of the students are into physics not because of the employment opportunities it offers. Both females and males agreed that they are studying physics because of job opportunities in physics.

Moreover, over 80% of all the age groups except those within 30–34, agreed with the statement. Of those within the ages 30–34, only 2 were not professional teachers (Table 9). This means that the remaining 33 students were professional teachers and gainfully employed by Ghana Education Service (GES). Their primary goal in studying physics could not have been employment in GES but probably for professional development.

From this study, majority of students belief that, pursuing studies in physics holds the promise of enhanced job prospects in the future, as evidenced by its ranking as the second most significant factor on the Relative Importance Index. Given the prevalent trend of declining enrolment rates among physics students across various tertiary institutions and colleges of education (Erinosho 2013; Opare et al. 2018), it is unsurprising to observe a substantial proportion of the students in the study, opting to pursue physics studies with the aim of securing employment opportunities post-graduation.

The scarcity of physics teachers in many educational institutions nationwide further reinforces the attractiveness of a career in physics, as graduates are poised to capitalize on the abundant job openings within the field. These findings resonate with the assertions

Table 9. Cross tabulation of job prospects in physics and age.

		Age group				Total
		20–24	25–29	30–34	35 – above	
I chose to read physics because it has better job prospect	Strongly disagree	1	1	4	0	6
	Disagree	2	3	6	0	11
	Not sure	2	0	3	2	7
	Agree	13	7	12	5	37
	Strongly agree	12	12	10	6	40
Total		30	23	35	13	101

put forth by Hazari et al. (2010), wherein career expectations emerged as a pivotal determinant motivating students to select physics as their area of specialization.

Moreover, students perceive a career in physics as providing opportunities for professional growth and advancement, while also fostering social integration and supporting their future aspirations, as highlighted by Fullarton and Ainley (2000). In addition, they expect that studying physics will open up greater prospects for continued academic development, consistent with the findings reported by Osborne and Collins (2001).

I did not choose physics but the university admitted me to read physics

A cohort of 32 students, despite not initially applying to study physics, received admission offers from the University (Table 3). Conversely, 56 students actively applied for admission to the physics programme, while an additional 13 students remained uncertain regarding their selection of physics as their preferred course of study.

Among the subset of 32 students who did not initially apply to study physics but were granted admission, a relatively small proportion, specifically 7 individuals, expressed regret following their enrolment in the physics programme. However, despite these sentiments of regret, the majority of these students assessed their academic performance within the programme as above average, as indicated in Table 10.

Within the academic discourse, a subgroup of students who perceive their enrolment in the physics program as a result of university allocation rather than active choice are identified as ‘non-choosers’. This conceptualization is recognized in scholarly works by researchers such as Holmegaard, Madsen, and Ulriksen (2014), Meli et al. (2020) and Venville et al. (2013). The university’s admission procedures permit students to indicate preferences for up to three courses, potentially leading to admission into the physics programme even if it was not their initial selection. However, this aspect ranks relatively low among the factors influencing students’ decisions to pursue physics within the Department.

I chose to read physics because I wanted to easily secure study leave with pay from Ghana education service (GES)

In the examination of professional teachers, our inquiry aimed to discern whether the provision of study leave with pay significantly influenced their decision to pursue physics studies at the University, as outlined in Table 11. Among this cohort, a notable majority, comprising 62.1%, cited the desire for a consistent financial means to support their educational endeavours as the primary motivating factor for selecting physics as their

Table 10. Do you have any regrets studying physics?

		I did not choose physics but the University offered me admission to read physics					Total
		Strongly disagree	Disagree	Not sure	Agree	Strongly agree	
Do you have any regrets studying Physics?	Yes	11	0	1	7	0	19
	No	37	8	12	23	2	82
	Total	48	8	13	30	2	101

Table 11. I chose to read physics because I wanted to easily secure study leave.

		Frequency	Percentage
I chose to read physics because I wanted to easily secure study leave with pay from GES	Strongly disagree	8	13.8
	Disagree	7	12.1
	Not sure	7	12.1
	Agree	22	37.9
	Strongly agree	14	24.1
Total		58	100

field of study. Conversely, a minority representing 25.9% of teachers, despite being on study leave with pay, did not deem this factor as the principal rationale behind their decision to enrol in physics courses.

The imperative of staff professional development holds particular significance for organizations, particularly within the educational sector. In Ghana, the Ghana Education Service (GES) has implemented study leave with pay as an integral component of its staff development initiatives, aimed at facilitating educators in advancing their education and acquiring the requisite competencies for effective classroom engagement (Osei-Gyasi 2010).

Eligibility for study leave with pay necessitates that teachers of professional standing have a minimum of 3 years of teaching experience (G.E.S 2021). This qualification process is highly competitive, resulting in only a limited number of teachers securing sponsorship annually. Notably, GES has allocated 60% of the total quota for staff development to science courses, with a pronounced emphasis on physics as the foremost priority.

Unsurprisingly, a significant majority of professional teachers have expressed a strong inclination toward choosing physics as their academic focus, driven by the singular opportunity it affords them to secure study leave with pay from GES. Recognizing the pivotal role of such programmes in influencing educators' academic choices, it becomes imperative for additional stakeholders and education policymakers to contemplate the establishment of scholarship schemes and alternative avenues of financial support. Such initiatives would extend assistance to students aspiring to pursue physics at the tertiary level, thereby contributing to a more diversified and enriched educational landscape.

Interviews with students

An interview was conducted to deepening and contextualizing the findings from the questionnaire administered. It is important to note that, while the questionnaire captured broad trends and patterns in students' motivations for choosing physics at UEW, the interviews provided rich, detailed insights into the personal experiences, challenges, and decision-making processes behind those choices. By engaging directly with students, the researcher was able to uncover underlying motivations, such as how long academic gaps affected their confidence in physics, and how institutional policies like study leave with pay significantly influenced their program selection.

Ten students were randomly selected and interviewed individually. All interviewees were professional teachers except for one. This selection was based on their unique academic and life trajectories, which may not have been fully captured through the questionnaire responses alone. Some of the professional teachers had teaching

experience ranging from 5 to 6 years, and they had spent between 6 and 16 years after completing senior high school before enrolling at the university.

All the participants interviewed claimed that, the number of years spent after SHS before entering the university was inhibiting their current performance in physics. They intimated that they had forgotten most of the basic concepts in physics; since they did not apply much of that in their teaching as posited by Thorndike (1913). They taught classes from Kindergarten (KG) to Junior High School (JHS). Their WASSCE performance in Physics were somewhat better than their performance in E-Maths. About 50% of the students interviewed had the same grades in physics and E-Maths.

Majority of the participants interviewed said they opted to read physics because of their great interest in the course. Some developed their love for physics at SHS while others developed theirs at the colleges of education. This underscores the reason why Opare et al. (2018) intimated that the students interest in the subject should be develop at the early stages of their academic journey.

The narrative concerning one of the participants proved to be both intriguing and insightful. The individual in question pursued a course of study in Early Childhood Education at the College of Education and subsequently accrued 5 years of teaching experience within the realm of Kindergarten education. Despite her earnest endeavours, she encountered impediments in her pursuit of further education, having twice applied for study leave with pay, only to face rejection on both occasions.

In recounting her experience, she disclosed that an official from the Ghana Education Service intimated to her that she would likely secure the scholarship if she were to opt for a discipline in physics. Motivated by this suggestion, she acquiesced to the proposal and altered her academic trajectory accordingly. Consequently, upon conforming to this directive, she successfully obtained approval for study leave with pay. Study leave with pay therefore plays a crucial role in students' choice of physics at the University of Education, Winneba.

Another student who sat home for 2 years before gaining admission said: 'I just chose to read physics'. She did not have any reason for choosing physics. When asked why she did not choose Biology, she said, she was not comfortable with the 'Big words' in Biology. This seems like a unique case among the lot, although interest seems to play a role. Otherwise, records indicate that more students opted for biology than physics in the University (Taale 2010).

Though students were interested in the course, they faced some challenges in pursuing the programme. All the participants claimed there was too much pressure in studying physics. They equally complained about spending too much of their time in studying the course. Some of the students claim that they easily forget what they learn. They however attributed the number of years they have spent not doing physics as the major cause. According to them, the lecturers did not make the physics concepts practical enough for them to learn. The students also complained about taking too many courses outside the Department of Physics Education.

When students were asked to suggest what they thought would help them learn physics better, they mentioned that the lecturers should make physics more practical. They want to see many exhibitions and symposia organized by the Department to showcase the importance of physics to basic and SHSs.

These narratives offered real-world context to support theoretical interpretations drawn from Rational Choice Theory, Social Cognitive Career Theory, and Krumboltz's Planned Happenstance Theory. Moreover, the interviews revealed practical challenges faced by students such as content retention difficulties, course overload, and a lack of hands-on learning which might otherwise be underreported in surveys. These insights are crucial for informing curriculum reform, targeted academic support, and policy recommendations. In essence, the interviews served to validate, explain, and expand on the statistical findings, providing a more comprehensive understanding of the factors influencing students' choice to study physics at UEW.

Conclusion

This study sought to answer two key research questions: (1) What factors significantly influence undergraduate students' decisions to pursue physics at UEW? and (2) How do demographic variables moderate these factors?

The factors influencing students at the University of Education, Winneba (UEW) in their decision to study physics are multifaceted. Rather than being shaped by a single determinant, students' choices are influenced by a combination of interrelated factors. These include personal interest, perceptions of the subject's difficulty, career aspirations, mentorship experiences, and the impact of governmental and institutional policies.

Personal interest ($R^2 = 0.92$) in the subject emerged as the most significant factor influencing students' decision to pursue physics. The majority of respondents indicated that their choice was driven by a genuine passion and enthusiasm for the discipline. This intrinsic motivation was consistently observed across various age groups, professional backgrounds, and durations spent after completing high school. Students who expressed a strong interest in physics also reported higher levels of academic engagement and greater persistence in addressing the challenges associated with the subject.

In addition, some students reported choosing physics precisely because they perceived it as a difficult and intellectually demanding subject. While this may appear counter-intuitive, it aligns with the concept of 'desirable difficulty', wherein cognitive challenge serves as a source of intrinsic motivation. For these students, the rigorous nature of physics presents an opportunity to demonstrate competence and resilience, both to themselves and to others. This perception was particularly common among younger students and female respondents, suggesting that the ability to master a challenging subject like physics is associated with increased social recognition and academic prestige.

The influence of role models particularly physics teachers emerged as another critical factor in students' decision-making processes. A majority of respondents indicated that they were encouraged by role models to pursue physics. For these students, the presence of a dedicated and passionate teacher not only enhanced the relatability and enjoyment of the subject but also played a significant role in shaping their academic choices. This finding underscores the pivotal role of educators in inspiring students and influencing their career trajectories within the sciences.

Parental influence was found to play a relatively modest role in shaping the decision of younger students to study physics, with this influence diminishing further among older students. This contrasts with much of the existing literature from Europe and the United States, where undergraduate cohorts are typically younger and more responsive to

parental guidance. In this study, however, the majority of participants were professional teachers generally older and more established who tended to make academic choices independently and were largely not affected by parental expectations. These results highlight the importance of considering socio-cultural and demographic factors in understanding educational decision-making and suggest the need for policy frameworks and program designs that respect student autonomy while offering inclusive support for diverse pathways into physics at the tertiary level.

Furthermore, many students view a degree in physics as a strategic pathway to a stable and well-remunerated career, with the anticipation of external rewards serving as a strong motivational factor. Within this context, the rational choice theory is particularly applicable, as students weigh the perceived costs and benefits of their academic decisions. The high demand for physics educators in Ghana, alongside the existing shortage of professionals in the field, reinforces the perception of physics as a viable and advantageous career option.

In addition to the above, the desire to pursue physics is also mediated by demographic characteristics such as age, gender, years spent after senior high school, and professional background. These demographic differences provide important context for understanding the diversity in students' motivations. Younger students were primarily drawn to physics because of its perceived intellectual difficulty, which they regarded as a challenge to be overcome. In contrast, older students demonstrated a more pragmatic approach to their decision-making, prioritizing factors such as job security, opportunities for study leave, and long-term professional development over academic challenge or social prestige. This pattern suggests that with increased age and life experience, students tend to place greater emphasis on practical and career-oriented considerations.

Professional background, particularly among in-service teachers, significantly influenced students' decision-making processes. For these individuals, the opportunity to obtain study leave with pay from the Ghana Education Service (GES) emerged as a critical factor in their choice to pursue a degree in physics. Many professional teachers indicated that physics was selected specifically because it provided a dependable route to securing financial support while advancing their academic qualifications.

However, professional teachers frequently reported difficulties in coping with the demands of coursework and highlighted the need for more practical, supportive, and learner-centered teaching methods. Interview findings revealed that several of these individuals had been absent from formal academic study for over a decade, contributing to challenges in content retention and overall academic performance.

Recommendations

We recommend that, educational institutions in Ghana should adopt a multifaceted strategy to enhance student motivation to pursue physics at the tertiary level. This approach should include curriculum reforms that emphasize the practical application of physics concepts, the establishment of structured mentorship programs, and the provision of comprehensive career guidance services. Additionally, the Ghana Education Service (GES) should ensure that all senior high schools (SHSs) are equipped with well-resourced laboratories to facilitate effective teaching and learning. GES is also encouraged to organize regular refresher courses for in-service physics teachers to keep them updated on contemporary pedagogical approaches and advancements in the discipline.

Furthermore, it is recommended that future research explore how the factors influencing students' choice of physics evolve over time and investigate comparative trends across different tertiary institutions in Ghana. Such studies would provide deeper insights into the dynamics of physics education and inform more targeted policy interventions.

We assert that the implementation of targeted interventions designed to cultivate a robust culture of physics education in pre-tertiary institutions can effectively stimulate students' interest in the subject. This, in turn, is expected to contribute to increased enrolment in physics programs at the University of Education, Winneba (UEW), as well as other tertiary institutions across Ghana.

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Data availability statement

The data that support the findings of this paper are within the article.

Link to questionnaire: <https://forms.gle/3MsVb32Bkpqqrn8n6>.

References

- Adu-Gyamfi, S., C. O. Marfo, A. Y. Nyaaba, K. Amakye-Boateng, M. Abass, and H. T. Yartey. 2020. "Free Senior High School (Shs) and Quality University Education in Ghana: The Role of the University Teacher." *Journal of Educational and Social Research* 10 (5): 225–25. <https://doi.org/10.36941/jesr-2020-0101>.
- Antwi, V. 2013. *Interactive Teaching of Mechanics in a Ghanaian University Context*. Utrecht, The Netherlands: Faculteit Bètawetenschappen Flsme, Utrecht University.
- Antwi, V., N. A. Sakyi-Hagan, F. Addo-Wuwer, and B. Asare. 2021. "Effect of Practical Work on Physics Learning Effectiveness: A Case of a Senior High School in Ghana." *East African Journal of Education and Social Sciences* 2 (3): 43–55. <https://doi.org/10.46606/eajess2021v02i03.0102>.
- Armstrong, P. I., S. X. Day, J. P. Mcvay, and J. Rounds. 2008. "Holland's RIASEC Model as an Integrative Framework for Individual Differences." *Journal of Counseling Psychology* 55 (1): 1. <https://doi.org/10.1037/0022-0167.55.1.1>.
- Arnold, J. 2004. "The Congruence Problem in John Holland's Theory of Vocational Decisions." *Journal of Occupational & Organizational Psychology* 77 (1): 95–113. <https://doi.org/10.1348/096317904322915937>.
- Bandura, A. 1986. "Social Foundations of Thought and Action." *Englewood Cliffs NJ* 1986 (23–28): 2.
- Bjork, E. L., and R. A. Bjork. 2011. "Making things hard on yourself, but in a good way: Creating desirable difficulties to enhance learning." In *Psychology and the real world: Essays illustrating fundamental contributions to society*, edited by M. A. Gernsbacher, R. W. Pew, L. M. Hough, and J. R. Pomerantz, 56–64. New York, NY: Worth Publishers.
- Bolshakova, V. L., C. C. Johnson, and C. M. Czerniak. 2011. "'It Depends on What Science Teacher You Got': Urban Science Self-Efficacy from Teacher and Student Voices." *Cultural Studies of Science Education* 6 (4): 961–997. <https://doi.org/10.1007/s11422-011-9346-2>.
- Bray, A., and J. Williams. 2020. "Why is Physics Hard? Unpacking students' Perceptions of Physics." *Journal of Physics: Conference Series*, 012002. IOP Publishing.
- Buabeng, I., and D. Ntow. 2010. "A Comparison Study of students' Reasons/Views for Choosing/Not Choosing Physics Between Undergraduate Female Non-Physics and Female Physics Students at University of Cape Coast." *International Journal of Research in Education* 2 (2): 44–53.
- Buabeng, I., F. D. Ntow, and C. D. Otami. 2020. "Teacher Education in Ghana: Policies and Practices." *Journal of Curriculum and Teaching* 9 (1): 86–95. <https://doi.org/10.5430/jct.v9n1p86>.
- Buabeng, I., T. A. Ossei-Anto, and J. G. Ampiah. 2014. "An Investigation into Physics Teaching in Senior High Schools." *World Journal of Education* 4 (5). <https://doi.org/10.5430/wje.v4n5p40>.
- Checkley, D. 2010. *High School students' Perceptions of Physics*. Lethbridge, Alta: University of Lethbridge, Faculty of Education.
- Chen, C. P., and S. Lappano. 2023. "Career Counselling Considerations for Mothers Returning to Work." *Canadian Journal of Career Development* 22 (1): 53–62. <https://doi.org/10.53379/cjcd.2023.353>.
- Coffie, I. S., B. B. Frempong, and E. Appiah. 2020. "Teaching and Learning Physics in Senior High Schools in Ghana: The Challenges and the Way Forward." *Advances in Research* 21 (3): 35–42. <https://doi.org/10.9734/air/2020/v21i330192>.
- Creswell, J. W., and V. L. P. Clark. 2017. *Designing and Conducting Mixed Methods Research*. Thousand Oaks, CA: Sage publications.
- De Cassia Nakano, T., and R. Primi. 2024. "Professional Interests in Academic Giftedness and Regular Students: The Influence of Sex, Grade Level, and Group Variables." *European Journal of Special Education Research* 10 (6). <https://doi.org/10.46827/ejse.v10i6.5586>.
- Dewitt, J., J. Osborne, L. Archer, J. Dillon, B. Willis, and B. Wong. 2013. "Young Children's Aspirations in Science: The Unequivocal, the Uncertain and the Unthinkable." *International Journal of Science Education* 35 (6): 1037–1063. <https://doi.org/10.1080/09500693.2011.608197>.
- Edjah, K., E. Domey, and F. Ankomah. 2019. "Experiences of Level 100 Undergraduate Students in Developing Countries: A Case Study in University of Cape Coast, Ghana." *International Journal of Social Sciences & Educational Studies* 5 (4): 13.

- Erinosho, S. Y. 2013. "How Do Students Perceive the Difficulty of Physics in Secondary School? An Exploratory Study in Nigeria." *International Journal for Cross-Disciplinary Subjects in Education* 3 (Special 3): 1510–1515. <https://doi.org/10.20533/ijcdse.2042.6364.2013.0212>.
- Fullarton, S., and J. Ainley. 2000. *Subject Choice by Students in Year 12 in Australian Secondary Schools. Longitudinal Surveys of Australian Youth Research Report*. ERIC.
- Gabay-Egozi, L., Y. Shavit, and M. Yaish. 2010. "Curricular Choice: A Test of a Rational Choice Model of Education." *European Sociological Review* 26 (4): 447–463. <https://doi.org/10.1093/esr/jcp031>.
- Garza, C., and M. Landeck. 2004. "College Freshmen at Risk—Social Problems at Issue: An Exploratory Study of a Texas/Mexico Border Community College." *Social Science Quarterly* 85 (5): 1390–1400. <https://doi.org/10.1111/j.0038-4941.2004.00282.x>.
- G.E.S. 2021. "Study Leave with Pay 2021–222 Notice." https://ges.gov.gh/wp-content/uploads/2021/07/STUDY-LEAVE-WITH-PAY-2021_2022-NOTICE.pdf.
- Green, D., and I. Shapiro. 1994. *Pathologies of Rational Choice Theory: A Critique of Applications in Political Science*. New Haven, CT: Yale University Press.
- Hapgood, M., M. J. Angling, G. Attrill, M. Bisi, P. S. Cannon, C. Dyer, J. P. Eastwood, S. Elvidge, M. Gibbs, and R. A. Harrison. 2021. *Development of Space Weather Reasonable Worst-Case Scenarios for the Uk National Risk Assessment*. Hoboken, NJ: Wiley Online Library.
- Hastie, R., and R. M. Dawes. 2009. *Rational Choice in an Uncertain World: The Psychology of Judgment and Decision Making*. Thousand Oaks, CA: Sage Publications.
- Hazari, Z., G. Sonnet, P. M. Sadler, and M. C. Shanahan. 2010. "Connecting High School Physics Experiences, Outcome Expectations, Physics Identity, and Physics Career Choice: A Gender Study." *Journal of Research in Science Teaching* 47 (8): 978–1003. <https://doi.org/10.1002/tea.20363>.
- Hogan, R. 1982. "A Socioanalytic Theory of Personality." *Nebraska symposium on motivation*. Lincoln, NE: University of Nebraska Press.
- Holgate, S. A. 2023. "The Demand for Physics Skills in the Uk Workplace." *Physics World* 36 (10): 44. <https://doi.org/10.1088/2058-7058/36/10/34>.
- Holland, J. L. 1959. "A Theory of Vocational Choice." *Journal of Counseling Psychology* 6 (1): 35. <https://doi.org/10.1037/h0040767>.
- Holmegaard, H. T., L. M. Madsen, and L. Ulriksen. 2014. "To Choose or Not to Choose Science: Constructions of Desirable Identities Among Young People Considering a Stem Higher Education Programme." *International Journal of Science Education* 36 (2): 186–215. <https://doi.org/10.1080/09500693.2012.749362>.
- Jones, G., and M. Trippenbach. 2016. *Inspiring the Young to Study Physics*. Paris: OECD Publishing.
- Ker, L. A., J. Dewitt, J. F. Osborne, J. S. Dillon, B. Wong, and B. Willis. 2013. *Aspires Report: Young people's Science and Career Aspirations, Age 10–14*.
- Khatib, B. A., Y. S. Poh, and A. El-Shafie. 2020. "Delay Factors Management and Ranking for Reconstruction and Rehabilitation Projects Based on the Relative Importance Index (Rii)." *Sustainability* 12 (15): 6171. <https://doi.org/10.3390/su12156171>.
- Krumboltz, J. D. 2009. "The Happenstance Learning Theory." *Journal of Career Assessment* 17 (2): 135–154. <https://doi.org/10.1177/1069072708328861>.
- Kumar, S. 2016. "Parental Influence on Career Choice Traditionalism Among College Students in Selected Cities in Ethiopia." *International Journal of Psychology and Educational Studies* 3 (3): 23–30. <https://doi.org/10.17220/ijpes.2016.03.003>.
- Lent, R. W., and S. D. Brown. 2019. "Social Cognitive Career Theory at 25: Empirical Status of the Interest, Choice, and Performance Models." *Journal of Vocational Behavior* 115:103316. <https://doi.org/10.1016/j.jvb.2019.06.004>.
- Lent, R. W., S. D. Brown, and G. Hackett. 1994. "Toward a Unifying Social Cognitive Theory of Career and Academic Interest, Choice, and Performance." *Journal of Vocational Behavior* 45 (1): 79–122. <https://doi.org/10.1006/jvbe.1994.1027>.
- Lipps, J. H. 1999. "Beyond Reason: Science in the Mass Media." In *Evolution! Facts and Fallacies*, edited by J. W. Schopf, 71–90. San Diego, CA: Academic Press.

- Mahmud, M., and A. Adnan. 2023. "Career Self-Efficacy and Parent Career Behavior in the Career Choice of Community College Students." *Journal for ReAttach Therapy and Developmental Diversities* 6 (3s): 524–530.
- Majo, S. 2018. *Factors Influencing Poor Performance in Science Subjects in Secondary Schools in Shinyanga Municipality*. Munich, Germany: GRIN Verlag.
- Masnack, A. M., S. S. Valenti, B. D. Cox, and C. J. Osman. 2010. "A Multidimensional Scaling Analysis of students' Attitudes About Science Careers." *International Journal of Science Education* 32 (5): 653–667. <https://doi.org/10.1080/09500690902759053>.
- Meli, K., K. Lavidas, and D. Koliopoulos. 2020. "Factors That Influence Students in Choosing Physics Programmes at University Level: The Case of Greece." *Research in Science Education* 50 (3): 1075–1091. <https://doi.org/10.1007/s11165-018-9723-x>.
- Mellers, B. A., A. Schwartz, and A. D. Cooke. 1998. "Judgment and Decision Making." *Annual Review of Psychology* 49 (1): 447–477. <https://doi.org/10.1146/annurev.psych.49.1.447>.
- MoE. 2018. "Free Senior High School Policy." <https://moe.gov.gh/index.php/the-technical-vocational-and-skill-training/>.
- Morgan, M. A., and G. Aboagye. 2022. "Students' Interest in Physics by Gender, School Type and Programme of Study." *International Journal of Research and Innovation in Social Science* 6 (8): 591–601. <https://doi.org/10.47772/IJRISS.2022.6830>.
- Mujtaba, T., and M. J. Reiss. 2013. "Inequality in Experiences of Physics Education: Secondary School Girls' and Boys' Perceptions of Their Physics Education and Intentions to Continue with Physics After the Age of 16." *International Journal of Science Education* 35 (11): 1824–1845. <https://doi.org/10.1080/09500693.2012.762699>.
- Mujtaba, T., and M. J. Reiss. 2014. "A Survey of Psychological, Motivational, Family and Perceptions of Physics Education Factors That Explain 15-Year-Old students' aspirations to Study Physics in Post-Compulsory English Schools." *International Journal of Science and Mathematics Education* 12 (2): 371–393. <https://doi.org/10.1007/s10763-013-9404-1>.
- Ndalichako, J. L. 2014. "Students' Subject Choice in Secondary Schools in Tanzania: A Matter of students' Ability and Interests or Forced Circumstances?" *Open Journal of Social Sciences* 2 (8): 49. <https://doi.org/10.4236/jss.2014.28008>.
- Ogu, M. I. 2013. "Rational Choice Theory: Assumptions, Strengths and Greatest Weaknesses in Application Outside the Western Milieu Context." *Arabian Journal of Business and Management Review* 1 (3): 90–99. <https://doi.org/10.12816/0003628>.
- Olaosebikan, O. I., and A. Olusakin. 2014. Effects of Parental Influence on adolescents' Career Choice in Badagry Local Government Area of Lagos State, Nigeria.
- Opare, S. A., F. O. Manu, J. K. Ackah, and S. M. Akrosumah. 2018. "An Investigation into Teaching and Learning Materials (Tlms) Science Tutors Use to Assess Physics Lessons in the Colleges of Education in Ghana." *American Journal of Modern Physics and Application* 5 (4): 91–96.
- Ornek, F., W. R. Robinson, and M. P. Haugan. 2008. "What Makes Physics Difficult?" *International Journal of Environmental & Science Education* 3 (1): 30–34.
- Osborne, J., and S. Collins. 2001. "Pupils' Views of the Role and Value of the Science Curriculum: A Focus-Group Study." *International Journal of Science Education* 23 (5): 441–467. <https://doi.org/10.1080/09500690010006518>.
- Osei-Gyasi, S. 2010. *Study Leave with Pay and Human Resource Development in the Metropolitan Education Directorate of Ghana Education Service Kumasi*. Cape Coast, Ghana: University of Cape Coast.
- Panakaje, N., M. T. Pandavarakallu, S. R. Parvin, K. Niveditha, S. P. D. G. Shenoy, and R. Joyce Fernandes. 2024. "Decoding Destinations: Unraveling the Factors That Shape Career Choices in Commerce and Management." *Cogent Education* 11 (1): 2356395. <https://doi.org/10.1080/2331186X.2024.2356395>.
- Parola, A., and J. Marcionetti. 2022. "Career Decision-Making Difficulties and Life Satisfaction: The Role of Career-Related Parental Behaviors and Career Adaptability." *Journal of Career Development* 49 (4): 831–845. <https://doi.org/10.1177/0894845321995571>.

- Ramadhani, E., A. T. Jannah, and R. D. Putri. 2020. "Analysis of Holland Theory Career Guidance in Student Career Planning." *ENLIGHTEN: Jurnal Bimbingan Konseling Islam* 3 (1): 19–25. <https://doi.org/10.32505/enlighten.v3i1.1492>.
- Regan, E., and J. Dewitt. 2014. "Attitudes, Interest and Factors Influencing Stem Enrolment Behaviour: An Overview of Relevant Literature." In *Interest and Recruitment in Science: Recruitment, Retention and Gender Equity in Science, Technology and Mathematics Higher Education*, edited by EK Henriksen, J Dillon, and J Ryder, 63–88. Dordrecht: Springer Netherlands.
- Report, V.-C. S. 2020. *Vice-chancellor's Annual Report to the 53rd Congregation*. University of Cape Coast.
- Rocconi, L. M., X. Liu, and G. R. Pike. 2020. "The Impact of Person-Environment Fit on Grades, Perceived Gains, and Satisfaction: An Application of holland's Theory." *Higher Education* 80 (5): 857–874. <https://doi.org/10.1007/s10734-020-00519-0>.
- Siddiky, M. R., and S. Akter. 2021. "The students' Career Choice and Job Preparedness Strategies: A Social Environmental Perspective." *International Journal of Evaluation & Research in Education* 10 (2): 421–431. <https://doi.org/10.11591/ijere.v10i2.21086>.
- Taale, K. D. 2010. "Factors Influencing students' Physical Science Enrolment Decision at the University of Education, Winneba." *African Journal of Educational Studies in Mathematics and Sciences* 8 (1): 63–80. <https://doi.org/10.4314/ajesms.v8i1.69104>.
- Taale, K. D. 2011. "Parental and Society Influence on Physics students' Enrolment Decisions in the University of Education, Winneba, Ghana." *Journal of Education and Practice* 2 (4): 24–33.
- Thorndike, E. L. 1913. *The Psychology of Learning*. New York: Teachers college, Columbia University.
- Tzanakou, C. 2017. "Higher Education Systems and Institutions, Greece." *Encyclopedia of International Higher Education Systems and Institutions* 1–10. https://doi.org/10.1007/978-94-017-8905-9_254-1.
- U.E.W. University of education, winneba 28th graduation basic statistics. 2024. <https://uew.edu.gh/downloads/basic-statistics>.
- Vautero, J., A. D. Silva, and M. Do Céu Taveira. 2021. "Family Influence on undergraduates' Career Choice Implementation." *International Journal for Educational and Vocational Guidance* 21 (3): 551–570. <https://doi.org/10.1007/s10775-020-09453-0>.
- Venville, G., L. Rennie, C. Hanbury, and N. Longnecker. 2013. "Scientists Reflect on Why They Chose to Study Science." *Research in Science Education* 43 (6): 2207–2233. <https://doi.org/10.1007/s11165-013-9352-3>.
- WAEC. 2022. *West African Senior School Certificate Examinations (Wassce) Chief examiners' Reports*. Accra: WAEC Publishers.
- Yahaya, F., R. Gimba, and C. Gana. 2020. "Validity and Reliability of the Newly Developed Malay-Language Health Belief of Bloating (HB-Bloat) Scale." *International Journal of Environmental Research and Public Health* 17 (8). <https://doi.org/10.3390/ijerph17082773>.
- Zainudin, Z. N., L. W. Foon, Y. M. Yusop, W. N. W. Othman, E. M. Engku, and M. A. Kamarudin. 2024. "A Review on Application of holland's Riasec Theory in Educational Settings." *The International Journal of Academic Research in Business & Social Sciences* 14 (5): 395–410. <https://doi.org/10.6007/IJARBS/v14-i5/21397>.